

Aprendizado Profundo

Modelos Multimodais - CLIP

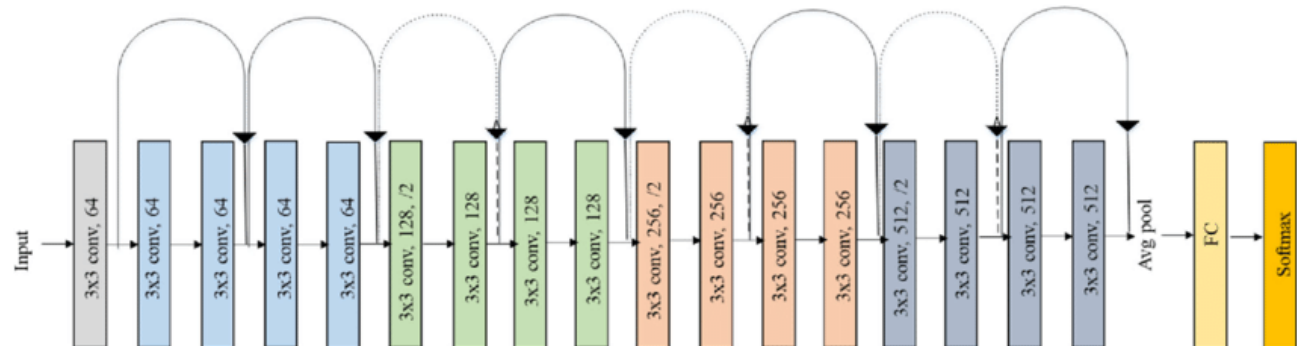
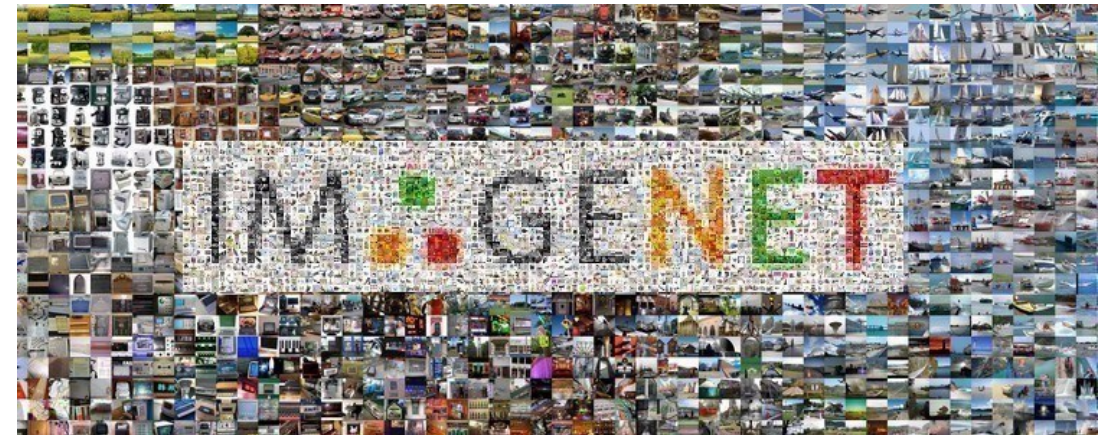
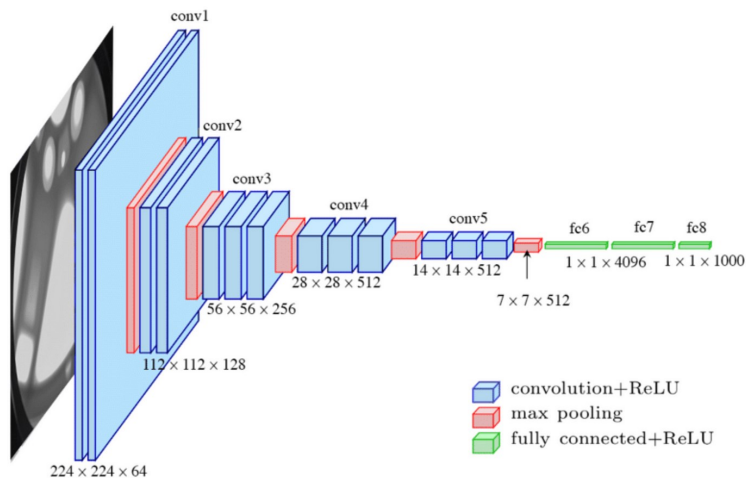
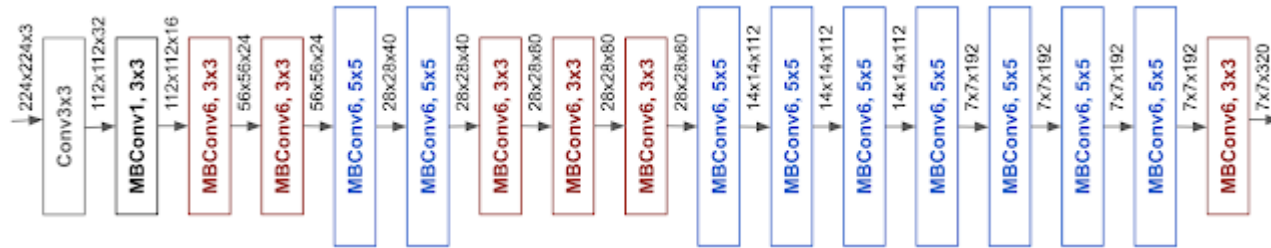
Professor: Lucas Silveira Kupssinskü

Agenda

- Contexto
- CLIP
 - Múltiplas modalidades
 - Diferentes Backends
 - Conjunto de Dados
 - Função de Custo
- Tarefas
- Limitações

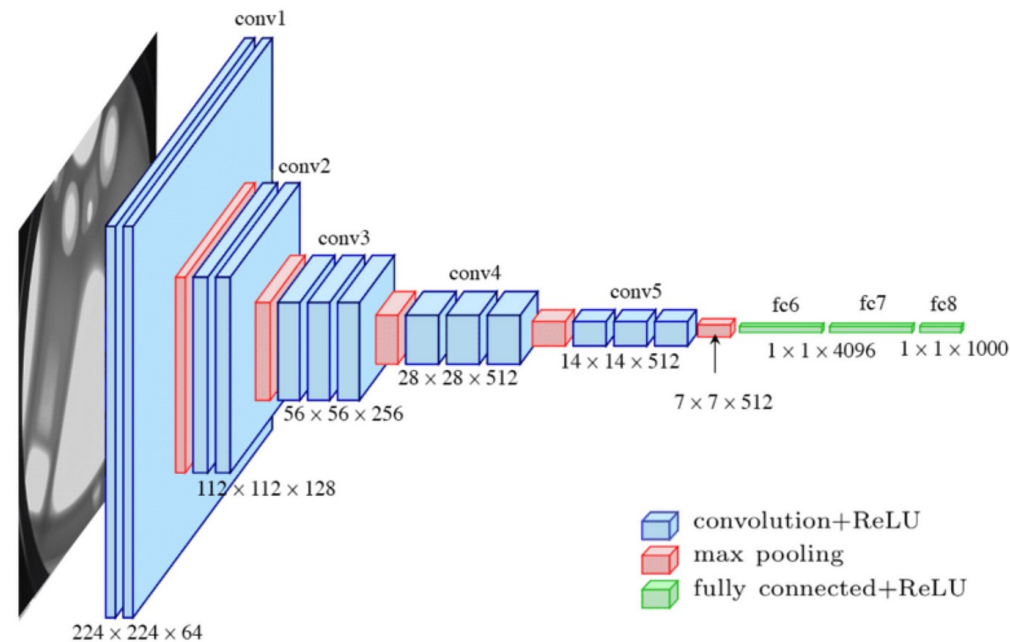
Contexto

- Uso de Transfer Learning para visão computacional já era uma realidade antes de 2021



Contexto

- Ideia típica era reaproveitar as camadas convolucionais e treinar apenas o classificador no final da rede
 - Ou fazer um treinamento de mais camadas com uma taxa de aprendizado pequena



Contexto

- Visão Computacional vs PLN
 - Enquanto PLN já estava se beneficiando do uso de datasets imensos em um setup de treinamento auto-supervisionado, a visão computacional ainda dependia de pré-treino em grandes conjuntos de dados manualmente anotados
 - O desafio é tentar criar uma forma de utilizar dados coletados da web em um processo de pré-treino sem necessidade de supervisão
 - Que favoreça Zero-shot para classes não conhecidas

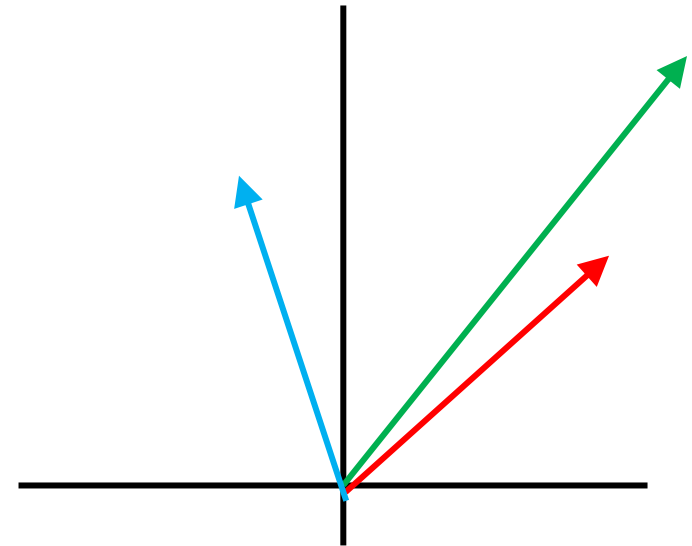
Contrastive Language-Image Pre-training

- Ao invés de fazer uma classificação em um conjunto de classes pré-definidas
- Vamos pegar um par (imagem, descrição) e aproximar os embeddings de cada uma das modalidades



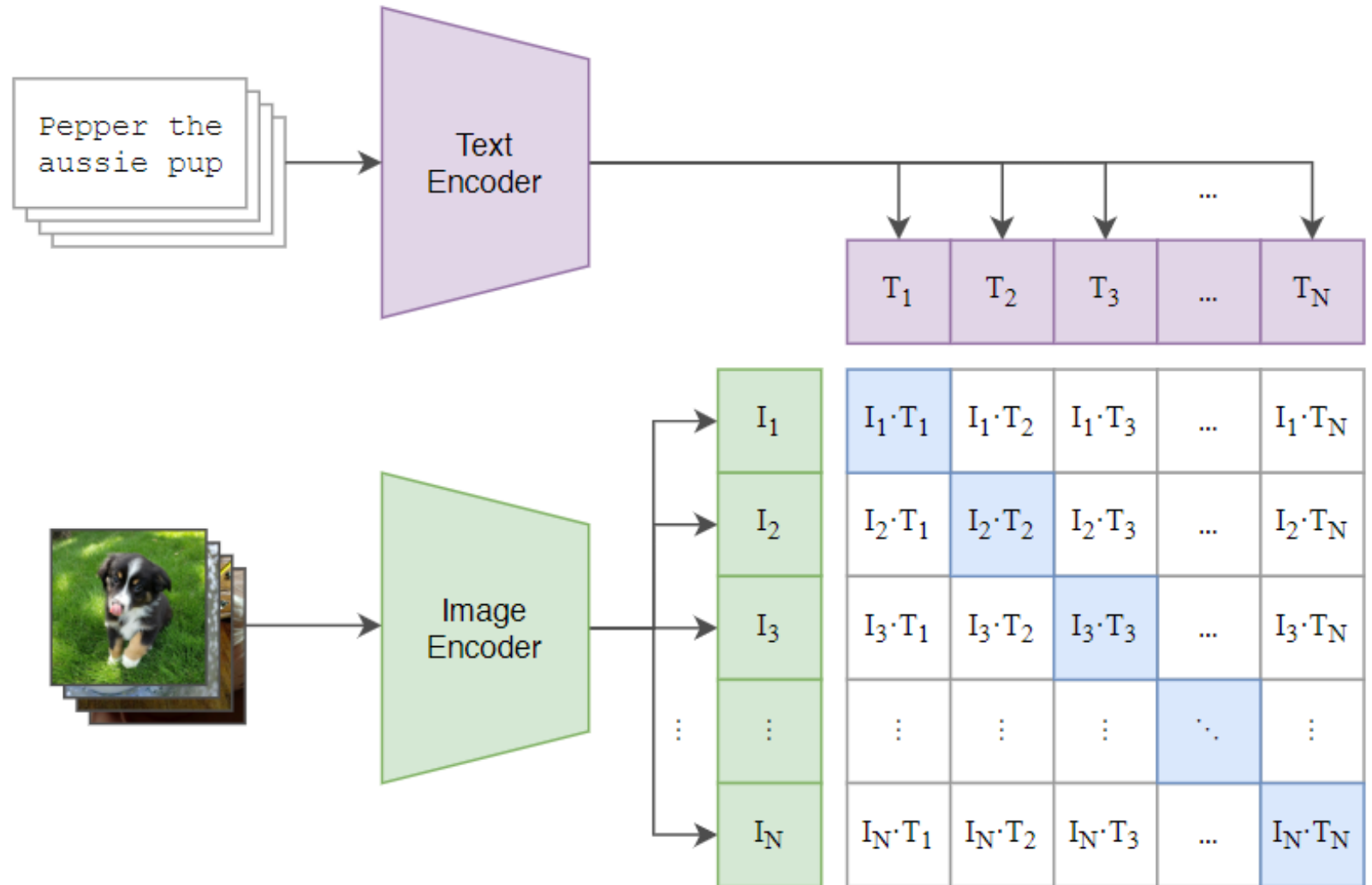
Um cachorrinho fofo

Um lobo feroz



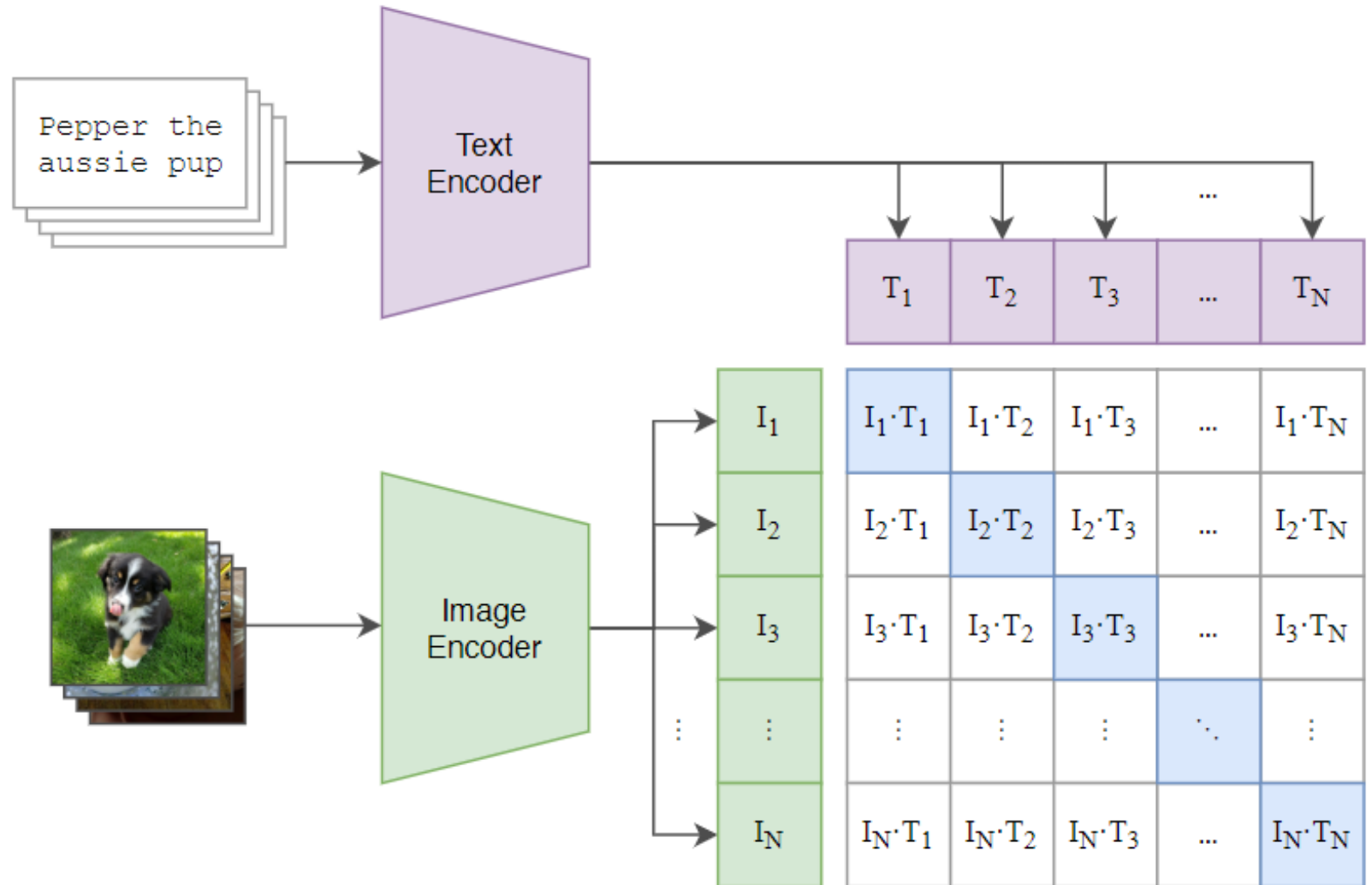
CLIP – *Contrastive Learning*

- O CLIP é formado um *encoder de texto* e um *encoder de imagem* que são treinados simultaneamente com objetivo de ter *embeddings similares*



CLIP – *Contrastive Learning*

- Computamos a similaridade do cosseno entre *embeddings* de texto e *embeddings* de imagem
- Em um *batch* de tamanho N vão existir N pares positivos para treinamento e $N^2 - N$ pares negativos



CLIP

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

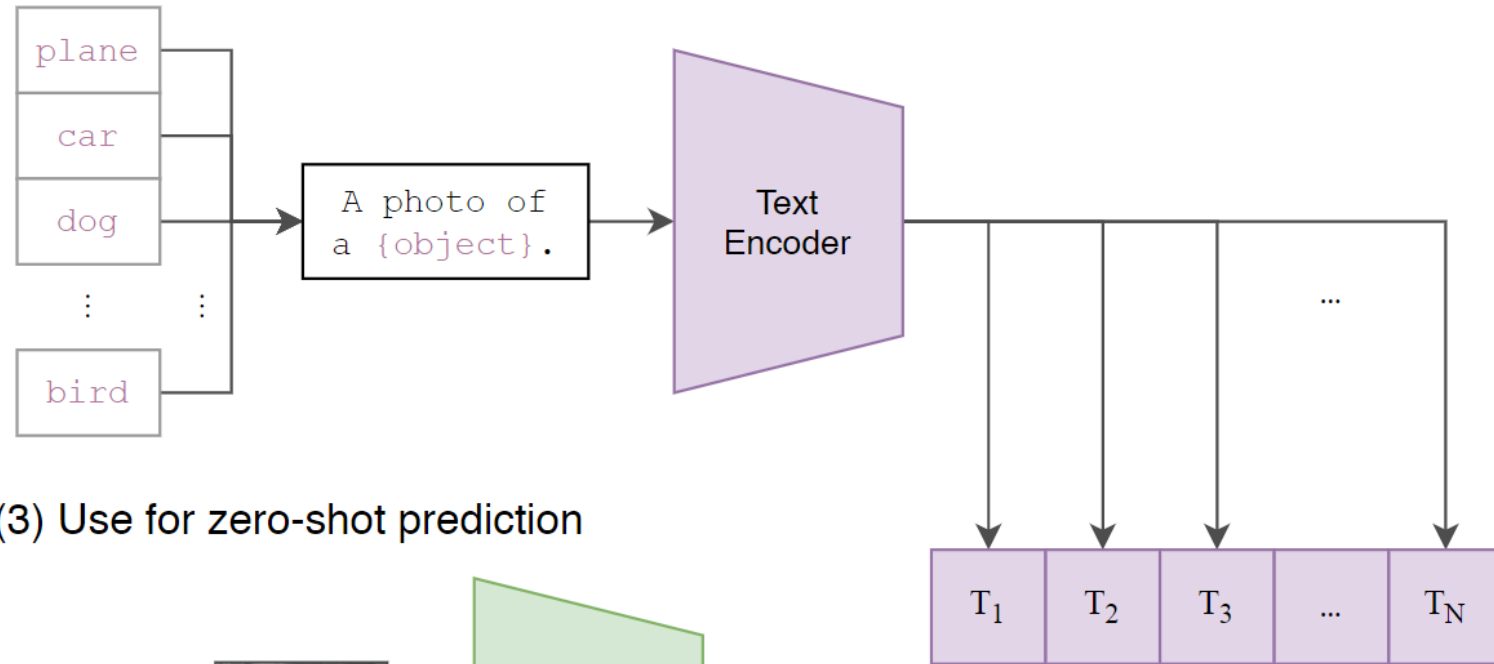
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

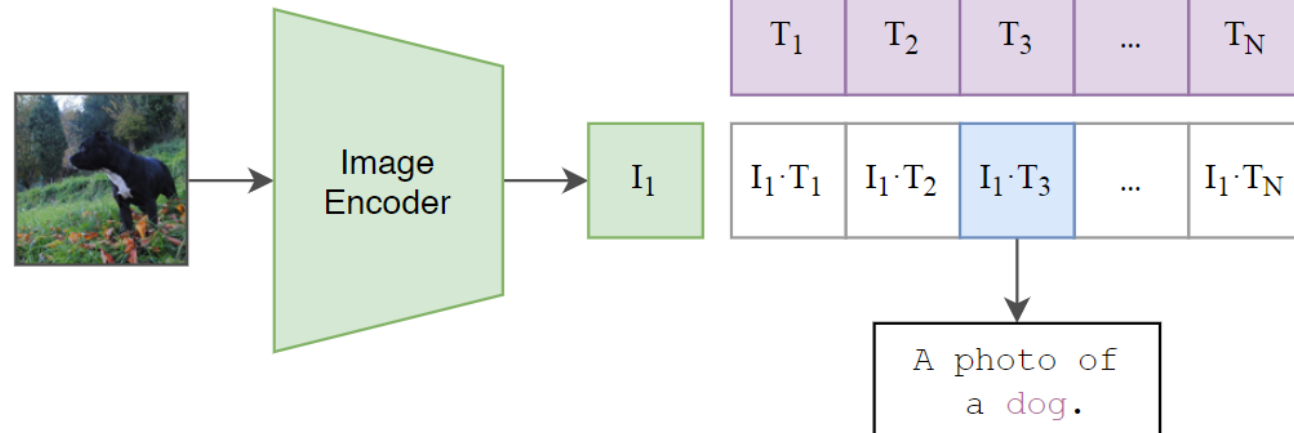
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

CLIP – *Contrastive Learning*

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



CLIP – Dataset e Treinamento

- Dataset fechado de 500M de pares (imagem, descrição)
- Hiperparâmetros definidos com “*uma mistura de*” *Grid Search, Random Search e Manual Tuning* na primeira época de treinamento
 - Treinamento
 - Com backend ResNet durou 18 dias em 592 GPUs V100
 - Com backend ViT-L/14 durou 12 dias em 256 GPUs V100

Hyperparameter	Value
Batch size	32768
Vocabulary size	49408
Training epochs	32
Maximum temperature	100.0
Weight decay	0.2
Warm-up iterations	2000
Adam β_1	0.9
Adam β_2	0.999 (ResNet), 0.98 (ViT)
Adam ϵ	10^{-8} (ResNet), 10^{-6} (ViT)

Model	Learning rate	Embedding dimension	Input resolution	ResNet blocks	width	layers	Text Transformer width	heads
RN50	5×10^{-4}	1024	224	(3, 4, 6, 3)	2048	12	512	8
RN101	5×10^{-4}	512	224	(3, 4, 23, 3)	2048	12	512	8
RN50x4	5×10^{-4}	640	288	(4, 6, 10, 6)	2560	12	640	10
RN50x16	4×10^{-4}	768	384	(6, 8, 18, 8)	3072	12	768	12
RN50x64	3.6×10^{-4}	1024	448	(3, 15, 36, 10)	4096	12	1024	16

Table 19. CLIP-ResNet hyperparameters

Model	Learning rate	Embedding dimension	Input resolution	Vision Transformer layers	width	heads	Text Transformer layers	width	heads
ViT-B/32	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-B/16	5×10^{-4}	512	224	12	768	12	12	512	8
ViT-L/14	4×10^{-4}	768	224	24	1024	16	12	768	12
ViT-L/14-336px	2×10^{-5}	768	336	24	1024	16	12	768	12

Table 20. CLIP-ViT hyperparameters

CLIP – Linear Probe Evaluation

- Para **avaliar a qualidade** dos *embeddings* utilizou-se um processo de *Linear Probe*
 - Treinar uma **Regressão Logística** usando os *embeddings* gerados em diversos datasets

Dataset	Classes	Train size	Test size	Evaluation metric
Food-101	102	75,750	25,250	accuracy
CIFAR-10	10	50,000	10,000	accuracy
CIFAR-100	100	50,000	10,000	accuracy
Birdsnap	500	42,283	2,149	accuracy
SUN397	397	19,850	19,850	accuracy
Stanford Cars	196	8,144	8,041	accuracy
FGVC Aircraft	100	6,667	3,333	mean per class
Pascal VOC 2007 Classification	20	5,011	4,952	11-point mAP
Describable Textures	47	3,760	1,880	accuracy
Oxford-IIIT Pets	37	3,680	3,669	mean per class
Caltech-101	102	3,060	6,085	mean-per-class
Oxford Flowers 102	102	2,040	6,149	mean per class
MNIST	10	60,000	10,000	accuracy
Facial Emotion Recognition 2013	8	32,140	3,574	accuracy
STL-10	10	1000	8000	accuracy
EuroSAT	10	10,000	5,000	accuracy
RESISC45	45	3,150	25,200	accuracy
GTSRB	43	26,640	12,630	accuracy
KITTI	4	6,770	711	accuracy
Country211	211	43,200	21,100	accuracy
PatchCamelyon	2	294,912	32,768	accuracy
UCF101	101	9,537	1,794	accuracy
Kinetics700	700	494,801	31,669	mean(top1, top5)
CLEVR Counts	8	2,000	500	accuracy
Hateful Memes	2	8,500	500	ROC AUC
Rendered SST2	2	7,792	1,821	accuracy
ImageNet	1000	1,281,167	50,000	accuracy

		Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Stanford Cars	FGVC Aircraft	VOC2007	DTD	Oxford Pets	Caltech101	Flowers102	MNIST	FER2013	STL10	EuroSAT	RESISC45	GTSRB	KITTI	Country211	PCam	UCF101	Kinetics700	CLEVR	HatefulMemes	Rendered SST2	ImageNet
CLIP-ResNet	RN50	81.1	75.6	41.6	32.6	59.6	55.8	19.3	82.1	41.7	85.4	82.1	65.9	66.6	42.2	94.3	41.1	54.2	35.2	42.2	16.1	57.6	63.6	43.5	20.3	59.7	56.9	59.6
	RN101	83.9	81.0	49.0	37.2	59.9	62.3	19.5	82.4	43.9	86.2	85.1	65.7	59.3	45.6	96.7	33.1	58.5	38.3	33.3	16.9	55.2	62.2	46.7	28.1	61.1	64.2	62.2
	RN50x4	86.8	79.2	48.9	41.6	62.7	67.9	24.6	83.0	49.3	88.1	86.0	68.0	75.2	51.1	96.4	35.0	59.2	35.7	26.0	20.2	57.5	65.5	49.0	17.0	58.3	66.6	65.8
	RN50x16	90.5	82.2	54.2	45.9	65.0	72.3	30.3	82.9	52.8	89.7	87.6	71.9	80.0	56.0	97.8	40.3	64.4	39.6	33.9	24.0	62.5	68.7	53.4	17.6	58.9	67.6	70.5
	RN50x64	91.8	86.8	61.3	48.9	66.9	76.0	35.6	83.8	53.4	93.4	90.6	77.3	90.8	61.0	98.3	59.4	69.7	47.9	33.2	29.6	65.0	74.1	56.8	27.5	62.1	70.7	73.6
CLIP-ViT	B/32	84.4	91.3	65.1	37.8	63.2	59.4	21.2	83.1	44.5	87.0	87.9	66.7	51.9	47.3	97.2	49.4	60.3	32.2	39.4	17.8	58.4	64.5	47.8	24.8	57.6	59.6	63.2
	B/16	89.2	91.6	68.7	39.1	65.2	65.6	27.1	83.9	46.0	88.9	89.3	70.4	56.0	52.7	98.2	54.1	65.5	43.3	44.0	23.3	48.1	69.8	52.4	23.4	61.7	59.8	68.6
	L/14	92.9	96.2	77.9	48.3	67.7	77.3	36.1	84.1	55.3	93.5	92.6	78.7	87.2	57.5	99.3	59.9	71.6	50.3	23.1	32.7	58.8	76.2	60.3	24.3	63.3	64.0	75.3
	L/14-336px	93.8	95.7	77.5	49.5	68.4	78.8	37.2	84.3	55.7	93.5	92.8	78.3	88.3	57.7	99.4	59.6	71.7	52.3	21.9	34.9	63.0	76.9	61.3	24.8	63.3	67.9	76.2

CLIP – Linear Probe Evaluation

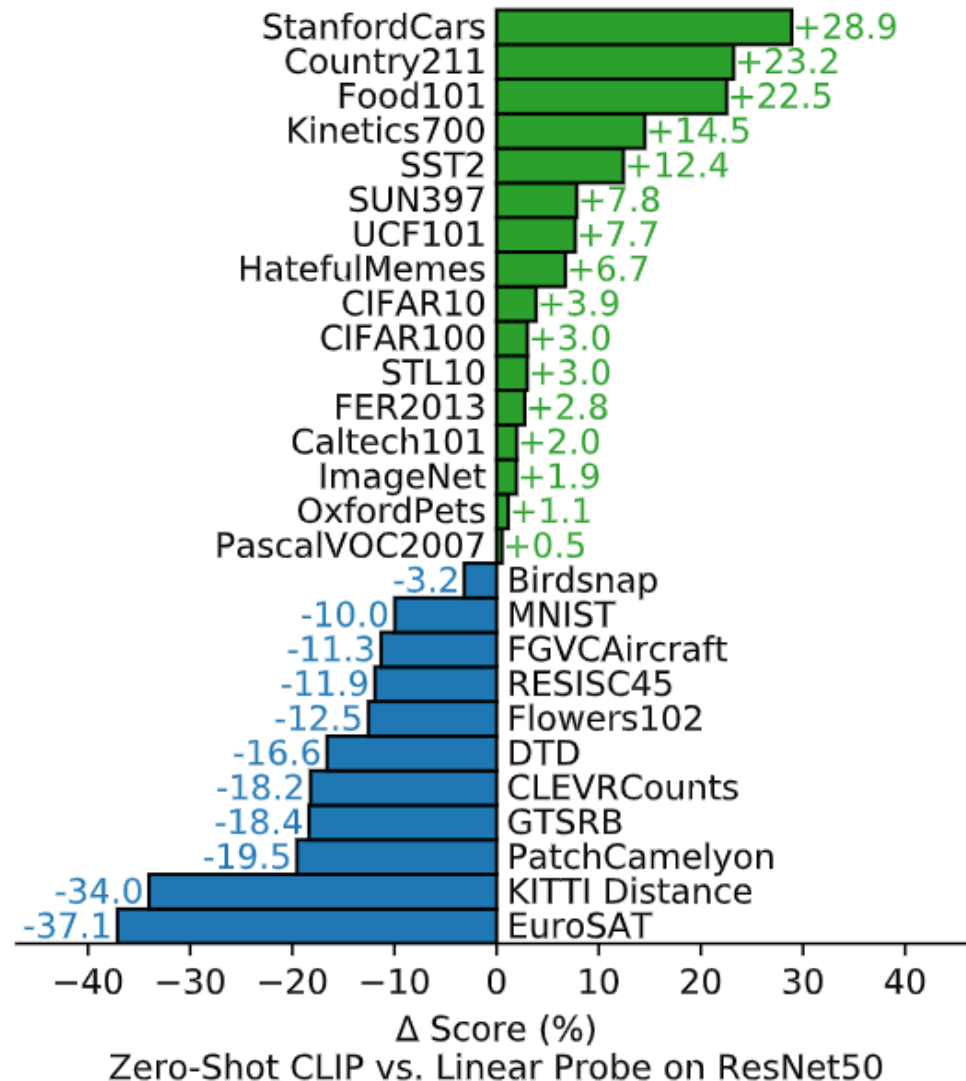


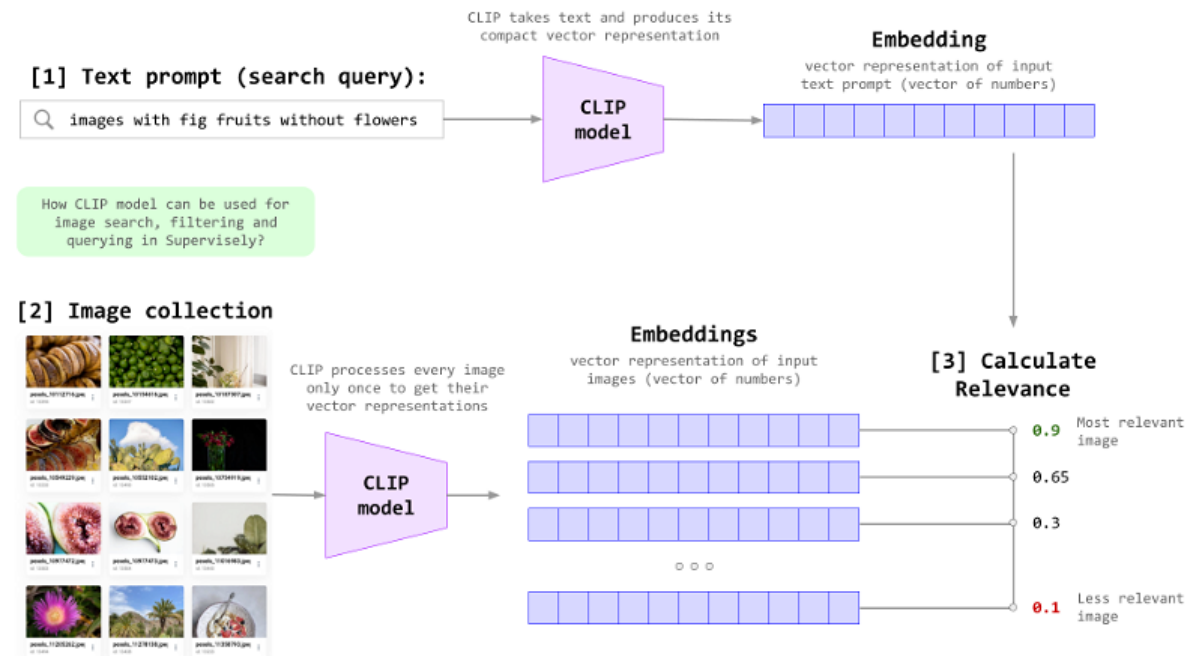
Figure 5. Zero-shot CLIP is competitive with a fully supervised baseline. Across a 27 dataset eval suite, a zero-shot CLIP classifier outperforms a fully supervised linear classifier fitted on ResNet-50 features on 16 datasets, including ImageNet.

Usos do CLIP

- *Zero-Shot Classification*
- *Fine Tuning Classification*: basta adicionar um classificador nos *embeddings* gerados e realizar fine tuning do classificador

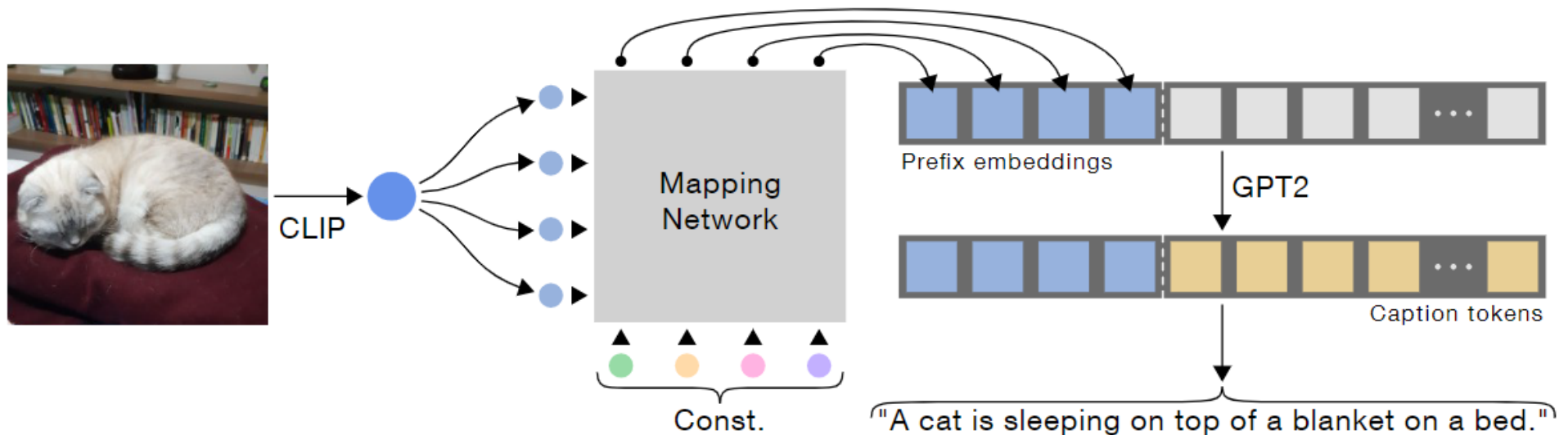
Usos do CLIP

- *Zero-Shot Classification*
- *Fine Tuning Classification*: basta adicionar um classificador nos *embeddings* gerados e realizar fine tuning do classificador
- *Semantic Retrieval*: Buscar imagens a partir de texto



Usos do CLIP

- *Zero-Shot Classification*
- *Fine Tuning Classification*: basta adicionar um classificador nos *embeddings* gerados e realizar fine tuning do classificador
- *Semantic Retrieval*: Buscar imagens a partir de texto
- *Captioning*: Gerar *captions* de imagens



Limitações

- É sensível a texto na imagem
- Generalização Zero-Shot é baixa para imagens diferentes do que consta no dataset (MNIST)
- Contagem de objetos na imagem
- Necessita de “*prompt engineering*” para algumas classificações



Granny Smith	85.6%
iPod	0.4%
library	0.0%
pizza	0.0%
toaster	0.0%
dough	0.1%



Granny Smith	0.1%
iPod	99.7%
library	0.0%
pizza	0.0%
toaster	0.0%
dough	0.0%

Outros modelos

- ALIGN: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision
- OpenCLIP: Reproducible scaling laws for contrastive language-image learning
 - Dataset LAION5B

Referências

- RADFORD, Alec et al. Learning transferable visual models from natural language supervision. In: **International conference on machine learning**. PMLR, 2021. p. 8748-8763.
- MOKADY, Ron; HERTZ, Amir; BERMANO, Amit H. Clipcap: Clip prefix for image captioning. **arXiv preprint arXiv:2111.09734**, 2021.
- SCHUHMANN, Christoph et al. Laion-5b: An open large-scale dataset for training next generation image-text models. **Advances in Neural Information Processing Systems**, v. 35, p. 25278-25294, 2022.
- <https://github.com/jacobmarks/awesome-clip-papers>